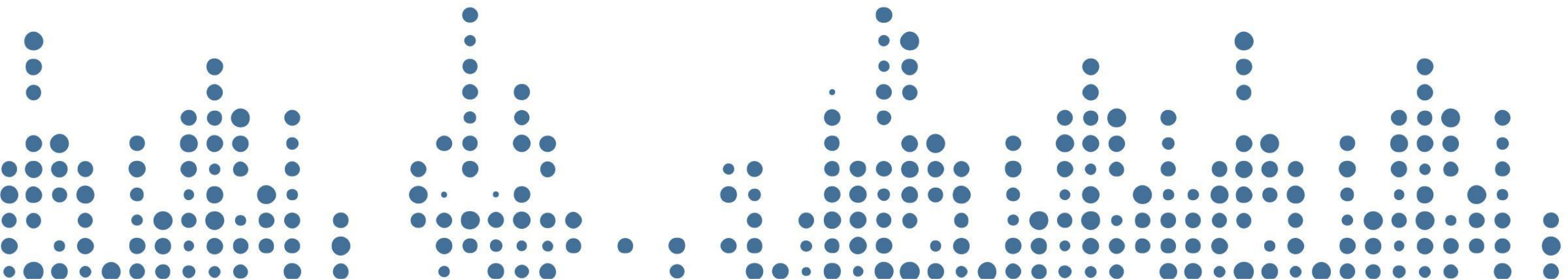




Getting Started with Positron

An AI-ready polyglot workflow for data science

`posit::conf(2026)`



Posit Academy

FREE

Self-paced, on-demand learning for every Posit user. No cost. Live now at academy.posit.co



Solutions

Products

Open Source

Customers

Resources

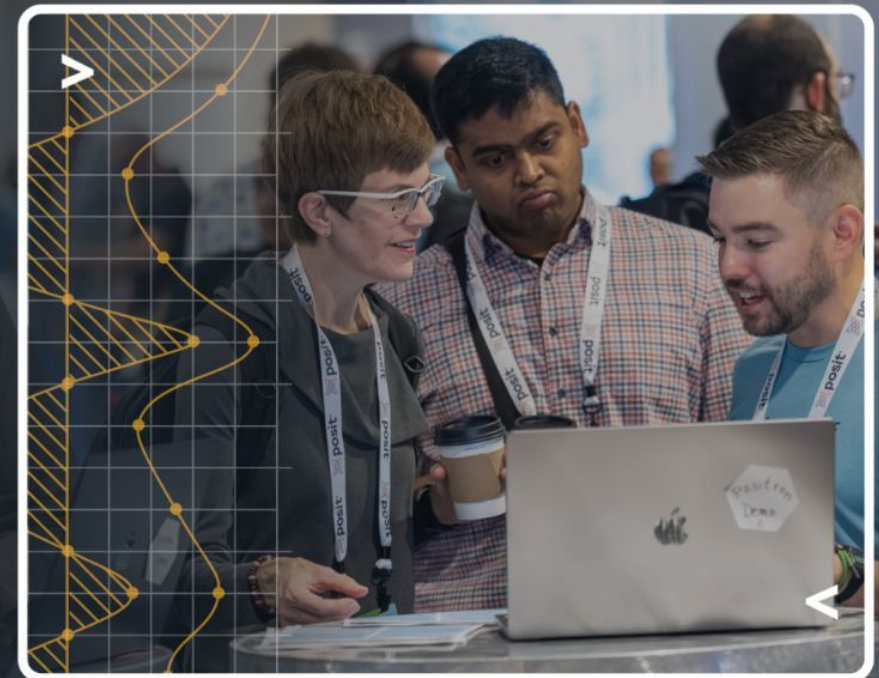
Search



Welcome to Posit Academy

Your learning hub for data science with R and Python — courses, live trainings, and a thriving community.

FULL COURSE CATALOG



Get Help

posit::conf(2026) is coming to Houston, TX! Join us Sept 14–16.

RSVP NOW

Your Turn

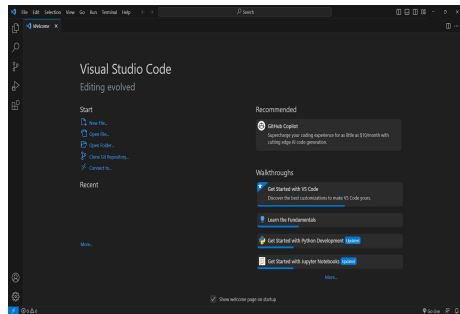
Poll - Which IDEs (Code Editors) have you used?

What is Positron ?

A free fork of Visual Studio Code purpose-built for **data science**

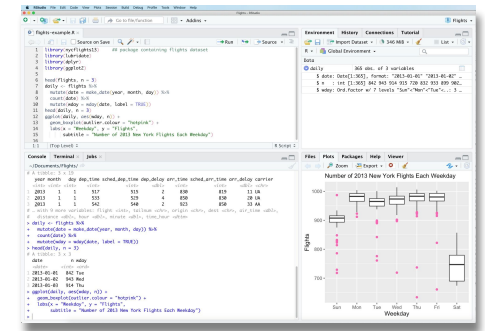
POSITRON

VS CODE

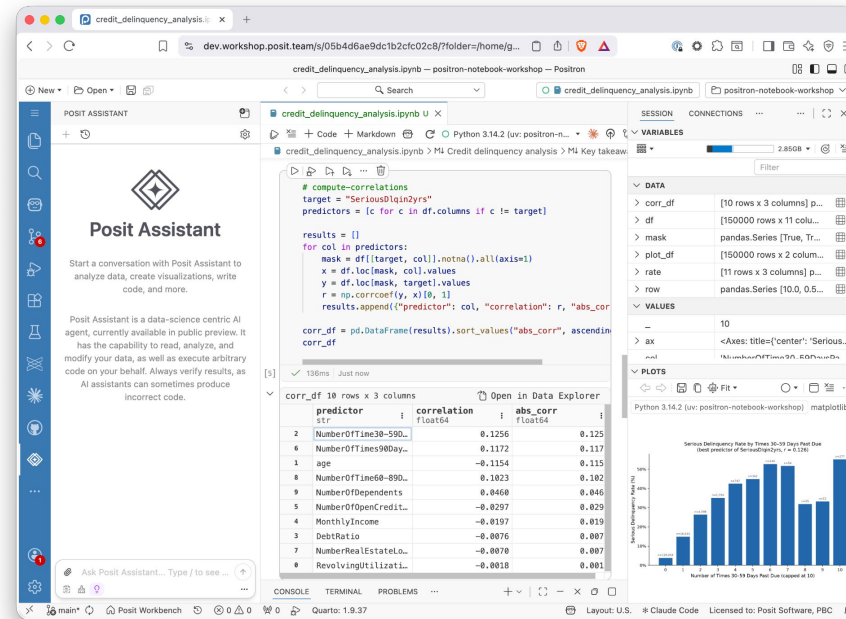


Software Developers

RSTUDIO

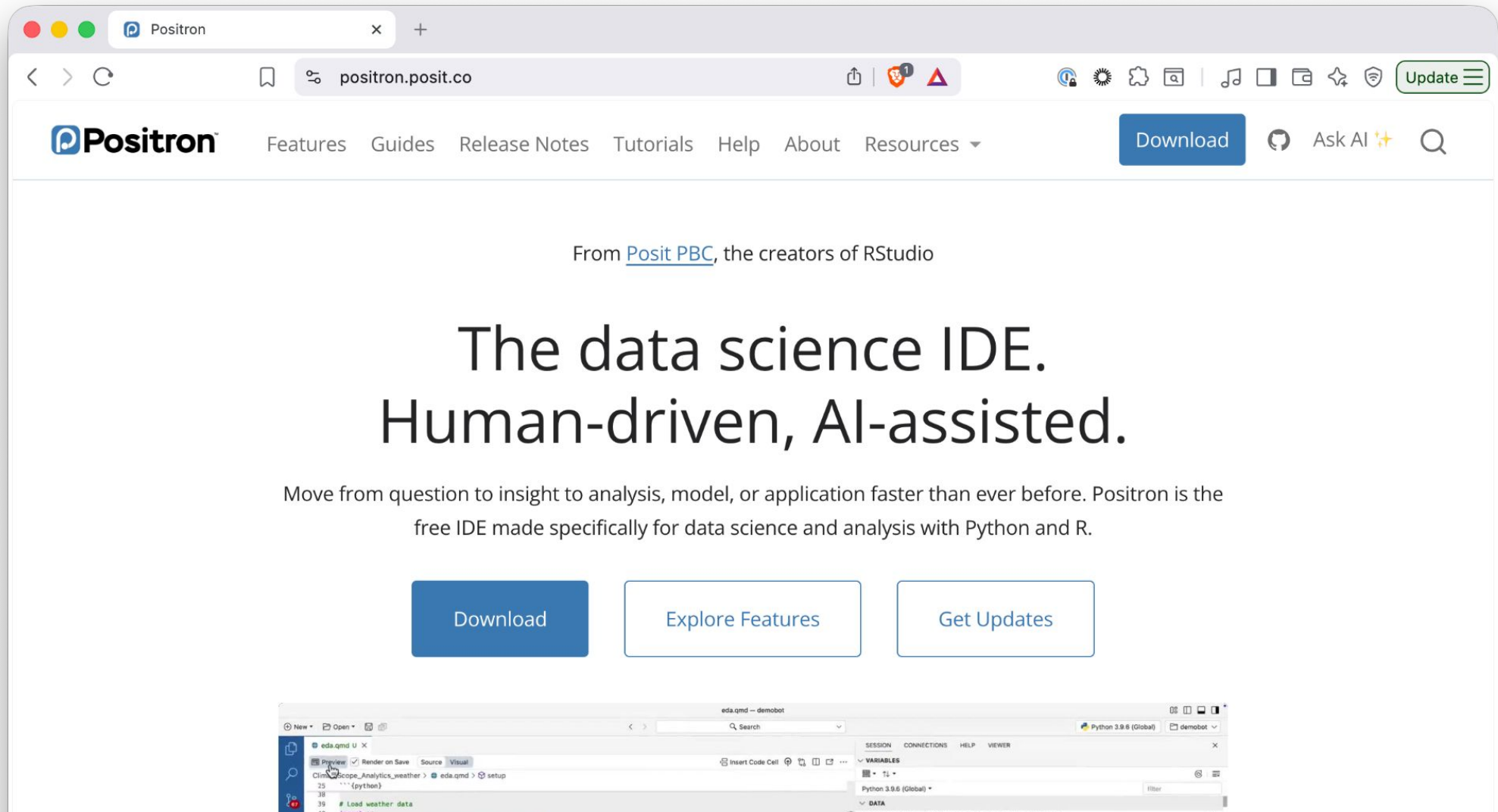


Data Scientists



Data Scientists

positron.posit.co



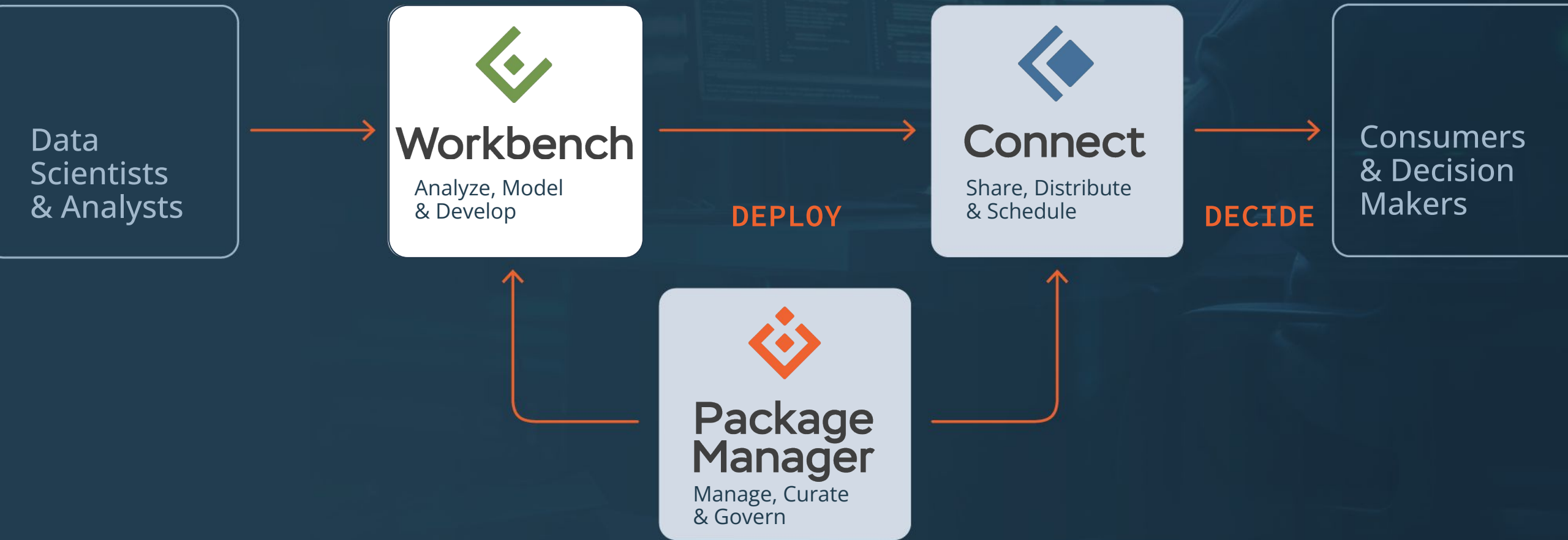
Your Turn

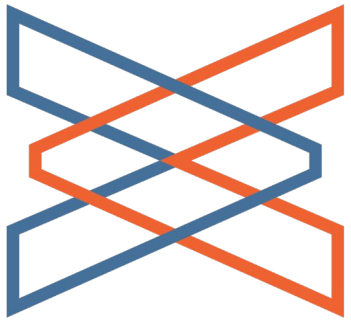
Poll - Which language do you use?

Learning Objective

Enable every participant to begin using Positron today for R or Python data science workflows.

We're hosting Positron on Workbench





posit®
formerly **R** Studio®



**OPEN
SOURCE
PLEDGE**
MEMBER



Posit's mission is to create free and open-source software for data science, scientific research, and technical communication, with the goal of enhancing the production and consumption of knowledge for everyone, regardless of economic means.



The Virtuous Cycle



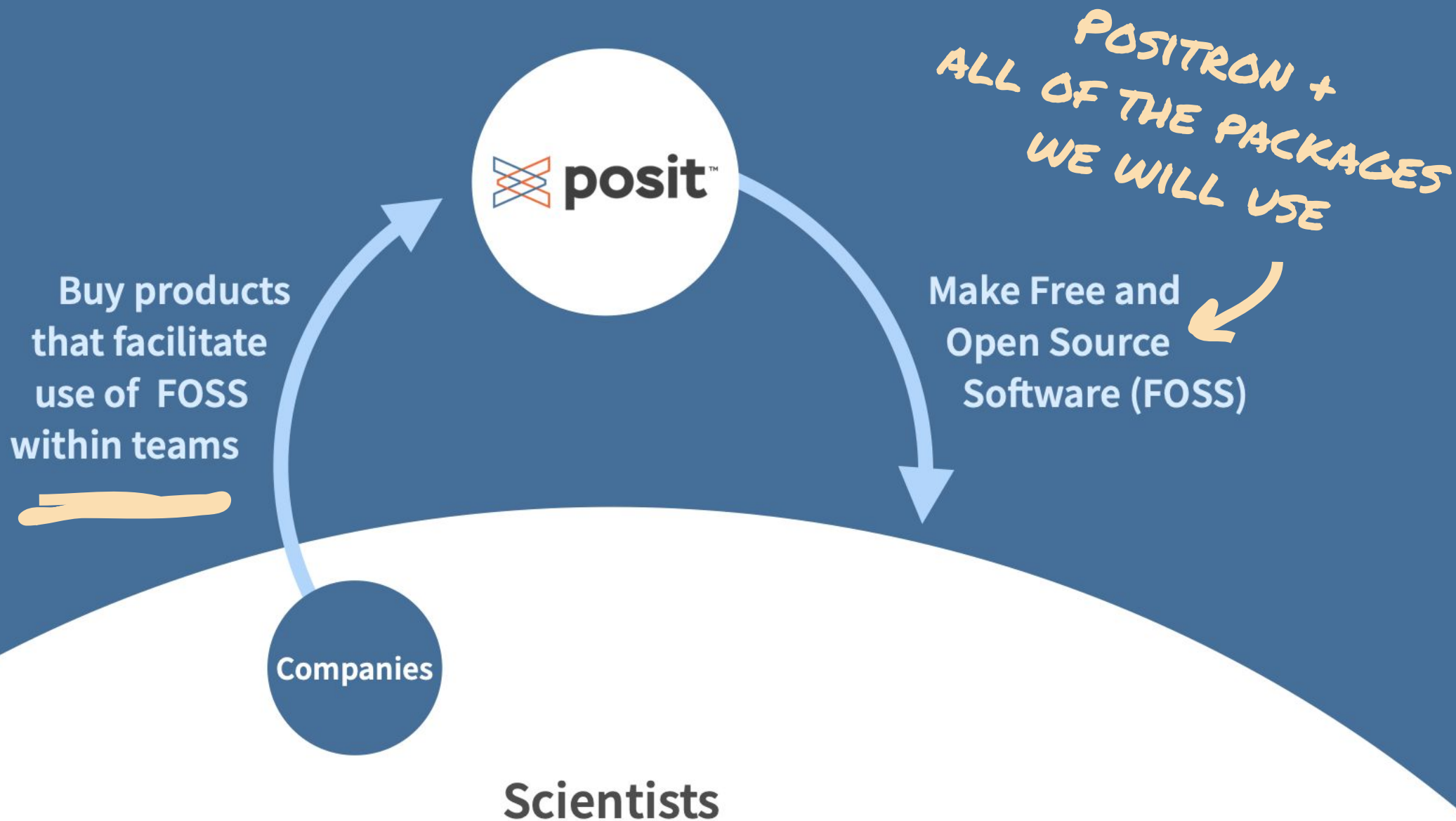
*POSITRON +
ALL OF THE PACKAGES
WE WILL USE*

Make Free and
Open Source
Software (FOSS)



Scientists

The Virtuous Cycle



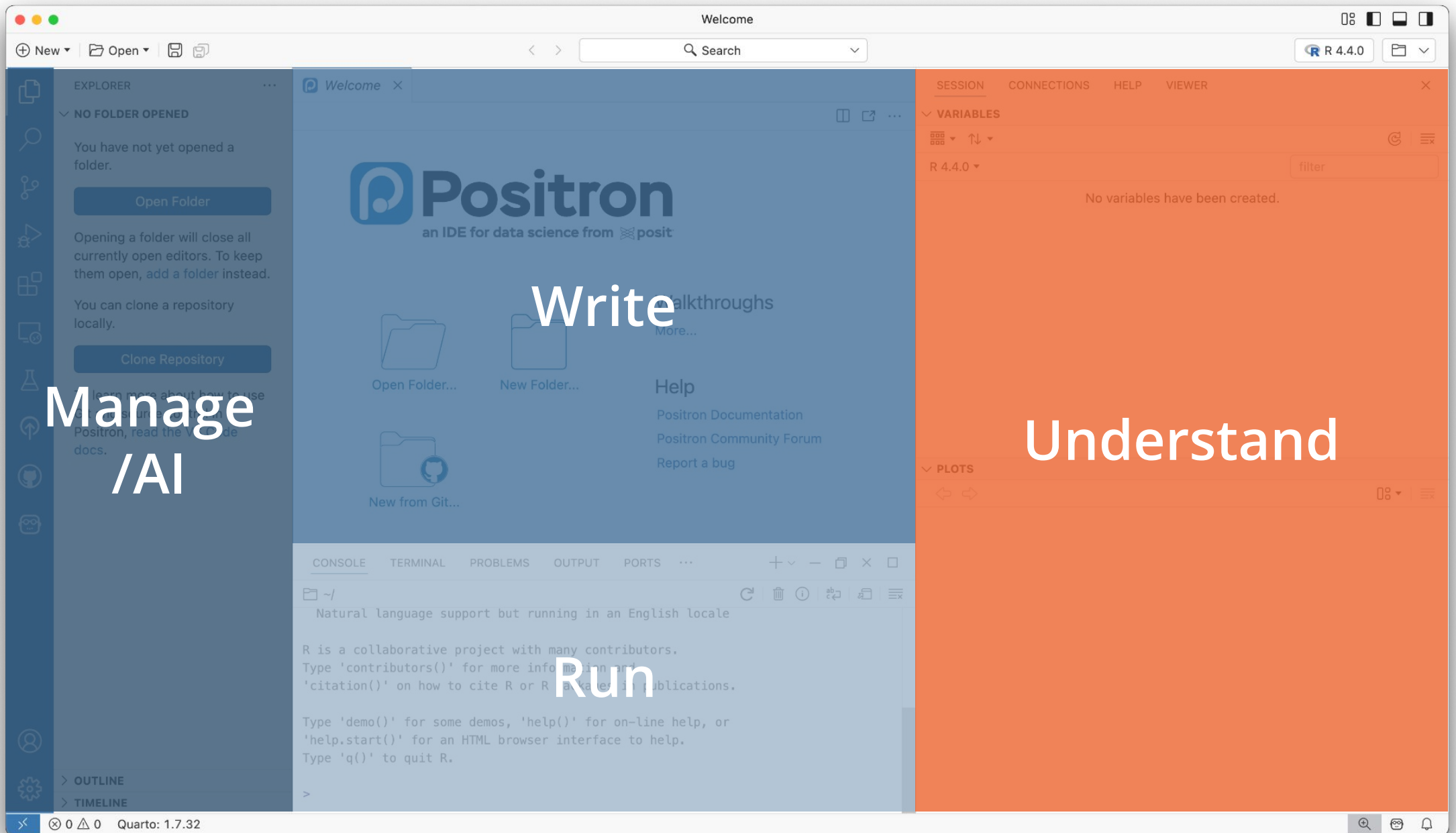
Hello Positron





Demo - Set up and tour of Positron





Projects = Folders

New Folder From Template

Python Environment

How would you like to set up your Python environment?

☒ Create a new virtual environment (Recommended)

☐ Use an existing environment

Environment Creation

Select a way to create a new virtual environment

uv Use uv to find a Python version locally (or install it if needed) for the new environment

Environment name

.venv

The environment will be created at

/Users/garrettgrolemond/Documents/posit/my-jupyter-notebook/.venv

Python Version

Select a Python version

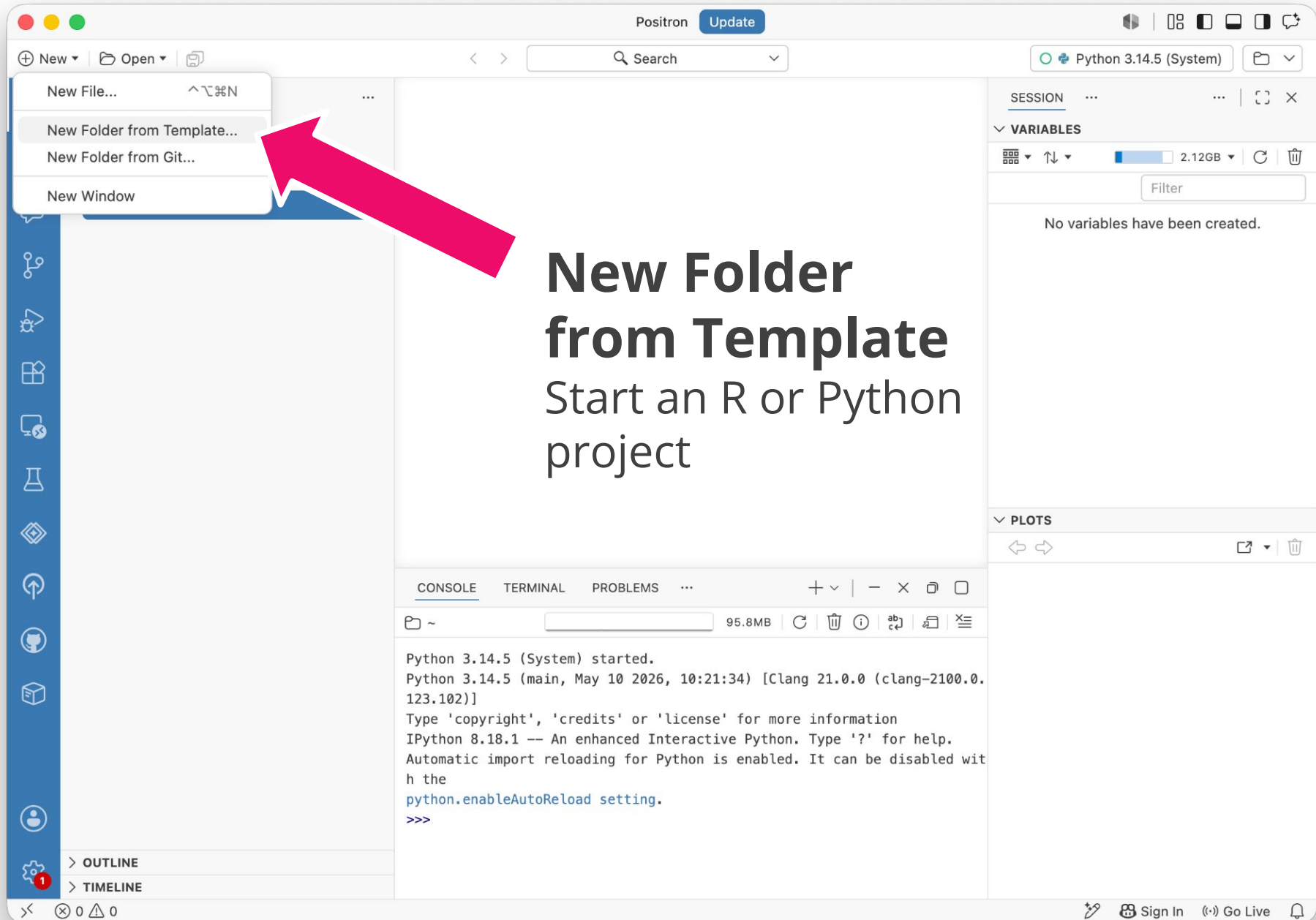
Python 3.14

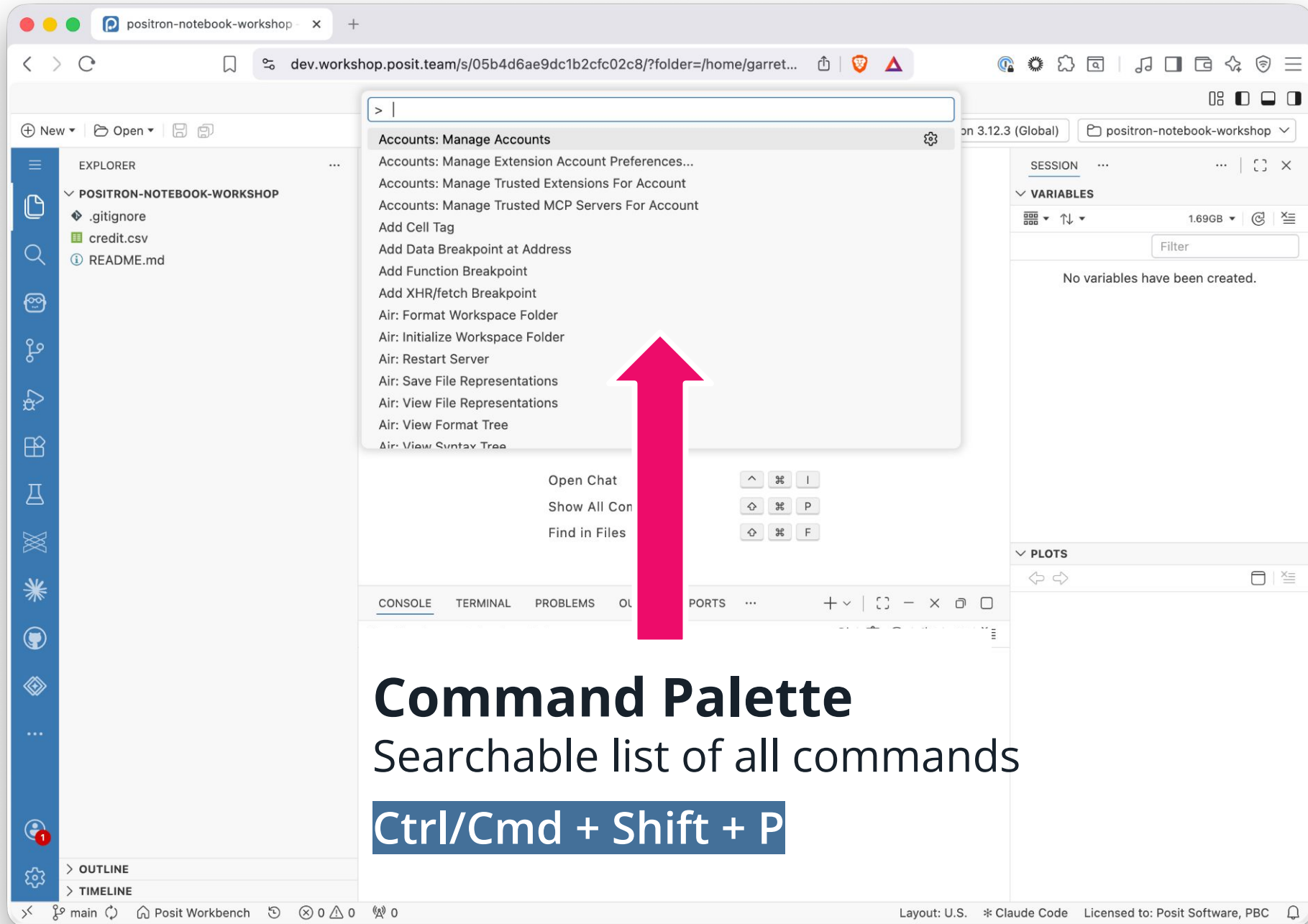
Back

Cancel

Create

Quickly set up directories and virtual environments with **New Folder from Template**





Your Turn

Open the Command Palette with
Cmd/Ctrl + Shift + P.

Find and run the command
Developer: Reload Window.

Your Turn

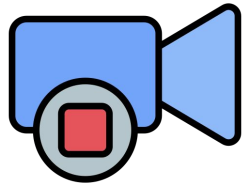
Open the Command Palette with
Cmd/Ctrl + Shift + P.

Find and run the command
Developer: Reload Window.

Today's materials



Will remain available after the workshop at
<https://github.com/posit-conf-2026/positron>

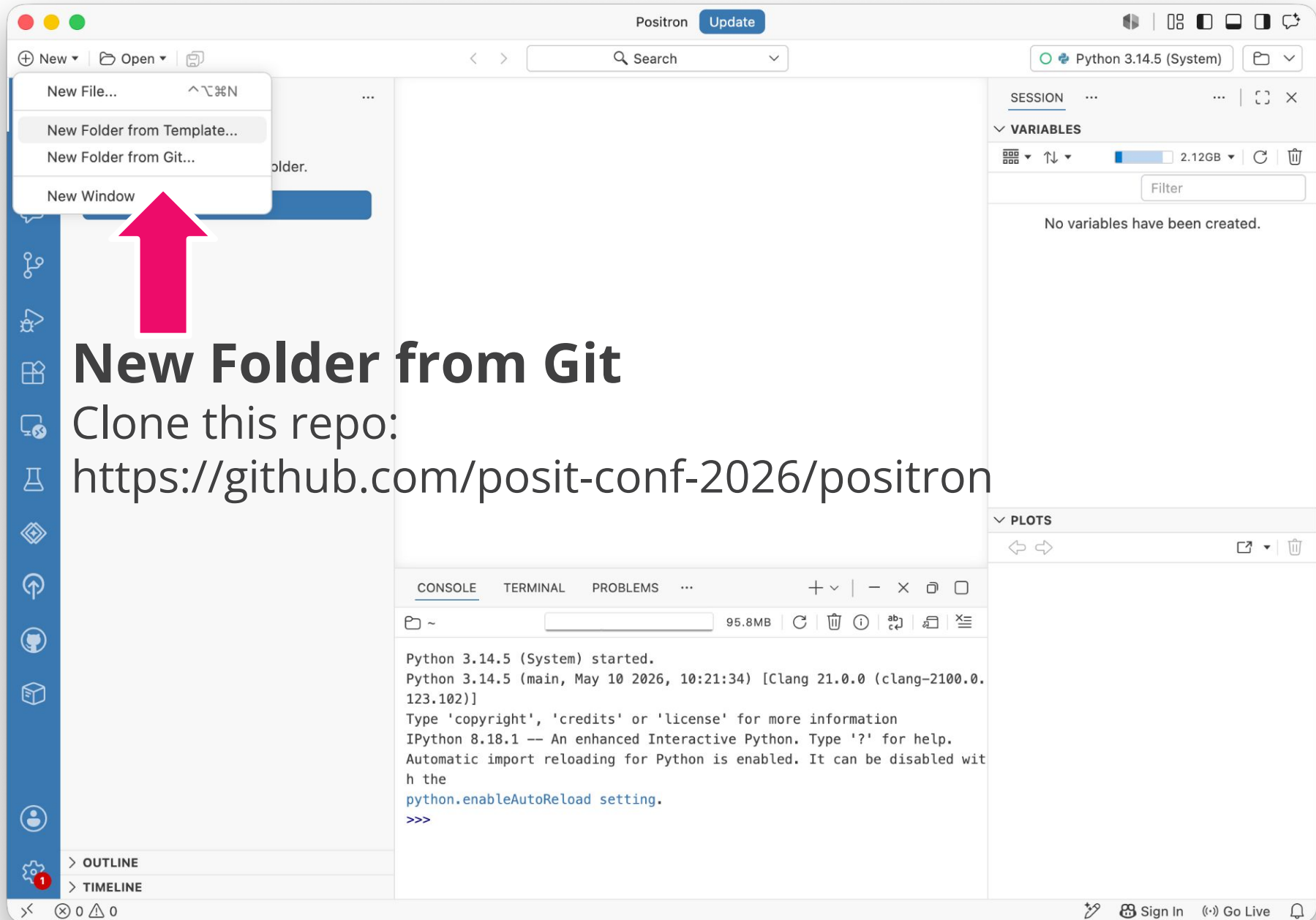


Recording available in a few days



Demo - Cloning a git repo and inspecting data





positron-notebook-workshop — Positron

Python 3.12.3 (Global)

positron-notebook-workshop

Layouts

Hide everything but the source pane

EXTENSIONS

Search Extensions in Marketplace

DEV.WORKSHOP.POSIT.TEAM - INSTALL... 17

- Air - R Language Support**
R formatter and language server
Posit Software, PBC
- Claude Code for VS C...** 802ms
Claude Code for VS Code: Harnes...
Anthropic
- Even Better TOML**
Fully-featured TOML support
tamasfe
- GitHub Pull Requests** 893ms
Pull Request and Issue Provider f...
GitHub
- Jupyter** 193ms
Jupyter notebook support, intera...
ms-toolsai
- Jupyter Cell Tags**
Jupyter Cell Tags support for VS ...
ms-toolsai
- Jupyter Keymap**
Jupyter keymaps for notebooks
ms-toolsai
- Jupyter Notebook Rend...** 2ms
Renderers for Jupyter Notebooks ...
ms-toolsai
- Jupyter Slide Show**
Jupyter Slide Show support for V...

BROWSER - INSTALLED 0

RECOMMENDED 0

SESSION ...

VARIABLES

No v... e been created.

PLOTS

CONSOLE TERMINAL PROBLEMS OUTPUT PORTS ...

~/positron-notebook-workshop

Python 3.12.3 (Global) started.
Python 3.12.3 (main, Apr 22 2025, 14:50:39) [GCC 11.4.0]
Type 'copyright', 'credits' or 'license' for more information
IPython 8.18.1 -- An enhanced Interactive Python. Type '?' for help.
Automatic import reloading for Python is enabled. It can be disabled with the [python.enableAutoReload](#) setting.
>>>

Layout: U.S. * Claude Code Licensed to: Posit Software, PBC

AI with Positron





Posit Assistant

An

AI client

within Positron.



Adapts *your LLMs* to data science. It:

1. Provides data science skills
2. Encourages a code-first, token saving method
3. Organizes access to Positron, R, and Python sessions

Power Positron with your preferred LLM

Configure Language Model Providers

Provider


Posit AI


Amazon Bedrock


Anthropic


Microsoft Foundry


OpenAI


Snowflake Cortex


GitLab Copilot
Pre

Authentication

☒ OAuth ☐ API Key

[Sign in](#)

Get started with Posit Assistant instantly via a free trial of [Posit AI](#), a managed service that provides access to frontier LLMs through a single account. Posit AI provides access to both Posit Assistant and Next Edit Suggestions.

By using Posit AI, you agree to the [Posit EULA](#), [Posit AI Terms of Service](#), and [Privacy Policy](#).

Your use of Posit AI is optional and at your sole risk.

[Close](#)



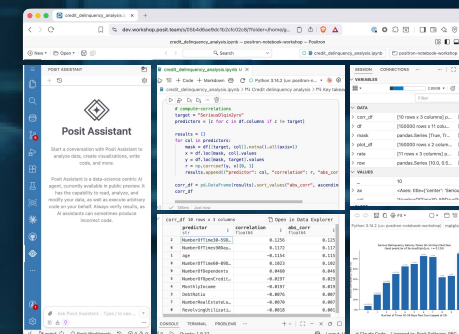
Demo - Open and configure Posit Assistant



AI



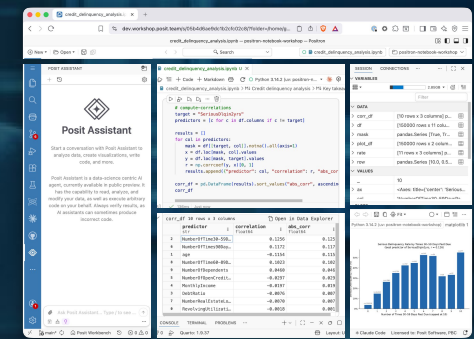
You





Posit Assistant

AI



You

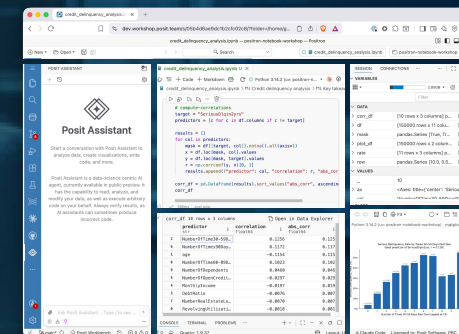


AI



Posit Assistant

?



Follow along with me

Open Posit Assistant ( in sidebar). Ask it to assess the data quality of `credit_raw.csv`, e.g:

Load the data in `credit_raw.csv`. What is the lowest age in the data set? The highest?

Follow along with me

Open Posit Assistant ( in sidebar). Ask it to assess the data quality of `credit_raw.csv`, e.g:

Load the data in `credit_raw.csv`. What is the lowest age in the data set? The highest?

You



AI



Posit Assistant

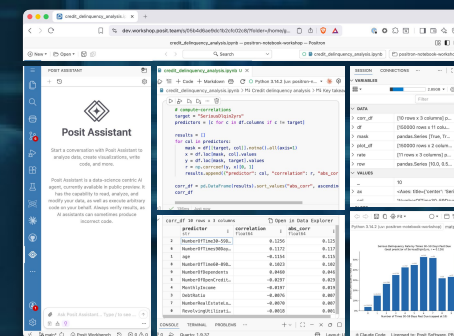
Skills, context



results

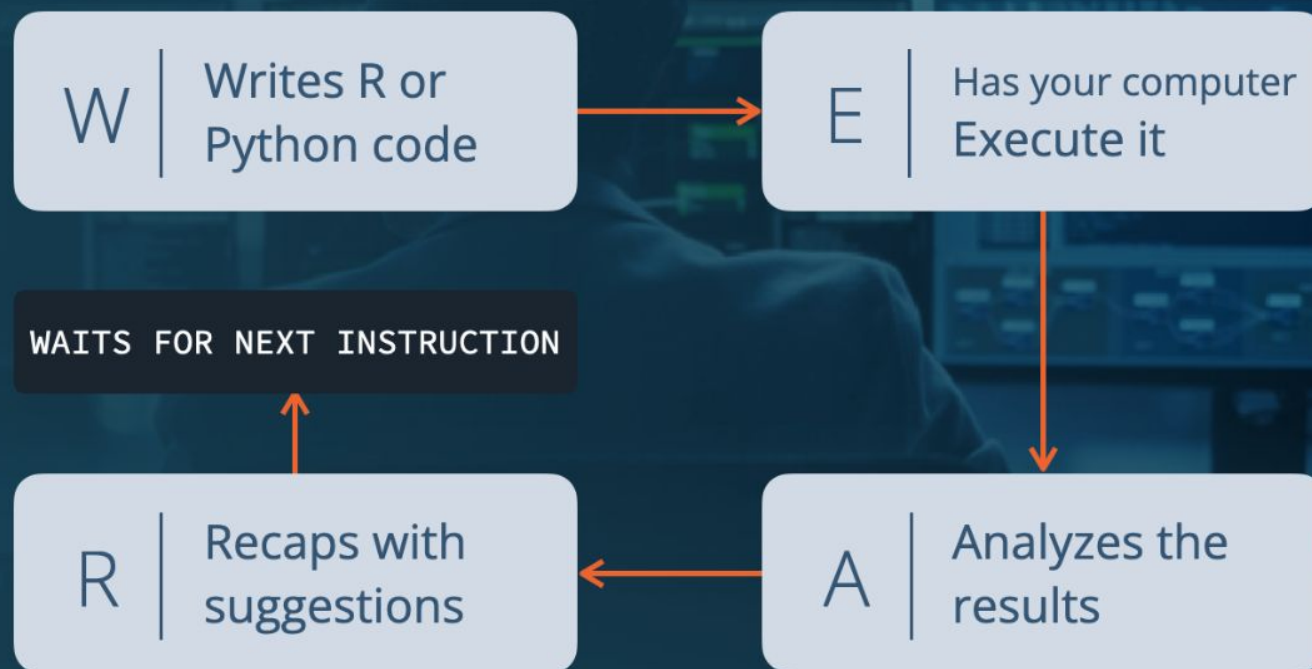


code



The
WEAR
philosophy

Code is the **most auditable**
way to explore data with AI



You



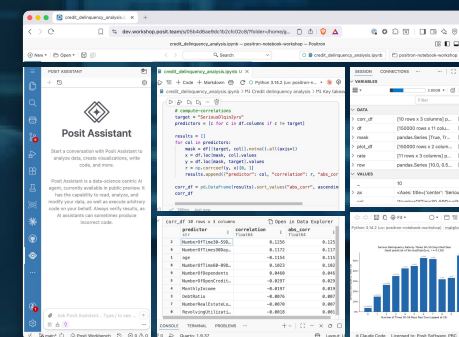
AI



Posit Assistant



code



Cleaning Data



Modes

Built in behavior modes for different types of tasks.



Assistant to analyze data, create visualizations, write code, and more.

Posit Assistant is a data-science centric AI agent. It has the capability to read, analyze, and modify your data, as well as execute arbitrary code.

-  **Normal**
Asks for approval for most edits.
-  **Plan mode**
In Plan mode, the assistant explores and designs an approach before making changes.
-  **Auto mode**
Evaluates actions for risk and seeks approval when appropriate.
-  **Cleaning mode**
In Cleaning mode, the assistant audits a dataset and surfaces cleaning decisions to you via the AskUser tool.

Your Turn

Open Posit Assistant ( in sidebar).

Choose **Cleaning mode**.

Ask Posit Assistant to:

Clean the data in credit_raw.csv. Save the clean version of the data as credit.csv.



Posit Assistant

Start a conversation with Posit Assistant to analyze data, create visualizations, write code, and more.

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In Plan mode, the assistant explores and designs an approach before making changes.



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Evaluates actions for risk and seeks approval when appropriate.



Cleaning mode

In Cleaning mode, the assistant audits a dataset and surfaces cleaning decisions to you via the AskUser tool.



Analyzing Data



Prompt Advice

For each prompt, explain:

1. **Why** - Explain your ultimate goal
2. **What** - Clear instructions for what you want it to do
3. **How** - Which functions, files, and *packages* you want it to use

I'd like to use the combined happiness dataset to understand how the USA's happiness score and ranking changed from 2015 to 2019. **Build a plot and a table to illustrate the changes.** Use the ggplot2 and gt packages to do this.

Your Turn

Clear the variables. Then ask Posit Assistant to build a predictive model, e.g:

I'm trying to understand the factors associated with loan delinquencies. Please load credit.csv. Then fit a simple logistic regression to the data that uses serious_dlqin_2yrs as a response variable. Use the stats (if R) / statsmodels (if Python) package.

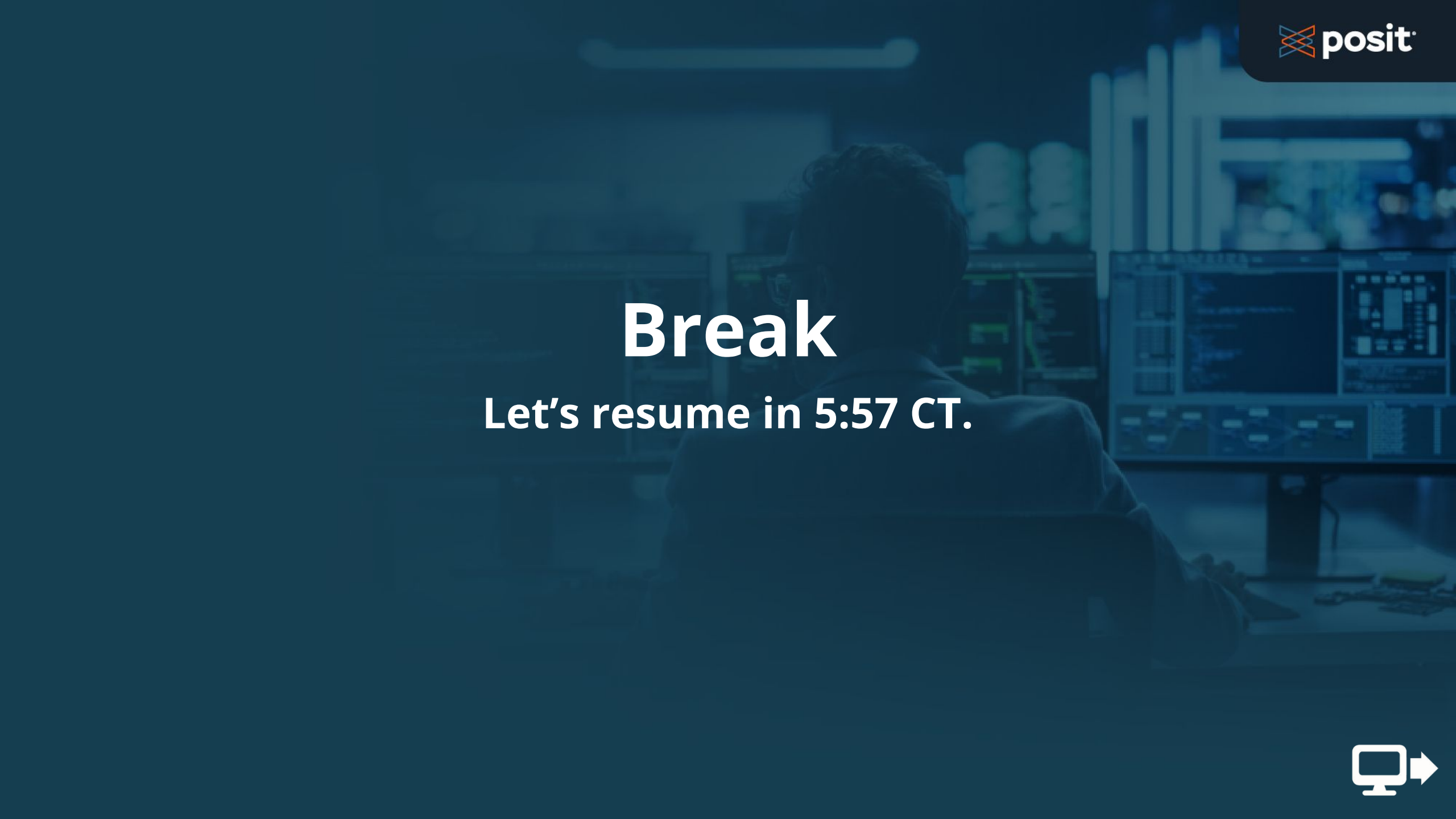
Follow up with your own queries until you understand what the model says.

Your Turn

Clear the variables. Then ask Posit Assistant to build a predictive model, e.g:

I'm trying to understand the factors associated with loan delinquencies. Please load credit.csv. Then fit a simple logistic regression to the data that uses serious_dlqin_2yrs as a response variable. Use the stats (if R) / statsmodels (if Python) package.

Follow up with your own queries until you understand what the model says.

The background is a dark, blue-tinted image of a person sitting at a desk with multiple computer monitors. The person is seen from the back, looking at the screens. The monitors display various data visualizations, including charts and graphs. The overall atmosphere is professional and tech-oriented.

Break

Let's resume in 5:57 CT.





Demo - Monitoring token usage in Posit Assistant, and working with AGENTS.md and skills.



Memory (AGENTS.md)

<https://assistant.posit.co/docs/features/memory/>

Skills

<https://assistant.posit.co/docs/features/skills/>

<https://github.com/posit-dev/skills>

Your Turn

Begin to explore the model results with visuals or tables. Start with one of the prompts below and then iterate with Posit Assistant's suggestions.

Draw an ROC curve for the model.

Use `gt` (R) or `great_tables` (Python) to make a table that lists the odds ratio for each predictor.

Use `ggplot2` (R) or `plotnine` (python) to make a calibration plot for the model.



Your Turn

Begin to explore the model results with visuals or tables. Start with one of the prompts below and then iterate with Posit Assistant's suggestions.

Draw an ROC curve for the model.

Use `gt` (R) or `great_tables` (Python) to make a table that lists the odds ratio for each predictor.

Use `ggplot2` (R) or `plotnine` (python) to make a calibration plot for the model.

Reporting results





Summarize your conversation
with the **/notebook** and
/report commands

Commands

/report Create a report as a Quarto document

/notebook Create a report as a Jupyter notebook

/



Claude 4 Sonnet ▾



Your Turn

Run `/report` or `/notebook` in Positron Assistant to save your code in an editable format.

Then,

- Click **Preview** at the top of the file (if quarto)
- Run all cells (if notebook)

Your Turn

Run `/report` or `/notebook` in Positron Assistant to save your code in an editable format.

Then,

- Click **Preview** at the top of the file (if quarto)
- Run all cells (if notebook)



Demo - Quarto reports for R and Python.



Notebooks with Positron



Notebooks...

retail_sales_fiscal_quarter.ipynb

Python 3.14.3 (uv: positron-n...)

retail_sales_fiscal_quarter.ipynb > M4 Retail sales by fiscal quarter > M4 Visualize quarterly sales > #| label:

Aggregate sales by fiscal quarter

```
#| label: aggregate-quarterly-sales
df["fiscal_quarter"] = df["order_date"].dt.to_period("Q").astype(str)

quarterly_sales = (
    df.groupby("fiscal_quarter", as_index=False)["sales"]
      .sum()
      .sort_values("fiscal_quarter")
)
quarterly_sales
```

[3] ✓ 64ms | 1 min ago

quarterly_sales 16 rows x 2 columns [Open in Data Explorer](#)

	fiscal_quarter str	sales float64
5	2016Q4	0.113.01
6	2016Q3	128560.21
7	2016Q4	180804.74
8	2017Q1	92686.37
9	2017Q2	135061.16
10	2017Q3	138056.37
11	2017Q4	234388.65
12	2018Q1	122260.88
13	2018Q2	127558.62
14	2018Q3	193815.84
15	2018Q4	278416.67

Visualize quarterly sales

Notebooks... with support

Publishing Assistant

SOURCE CONTROL

CHANGES

Adds notebook

Commit

Staged Changes

retail_sales_fiscal_quarter.ipynb

Changes

README.md

GRAPH

Moves data to a data frame

Adds to readme

Adds dataset

Initial commit

Package Manager

Packages

Filter packages

appprope 0.1.4

Disable App Nap on macOS >= 10.9

asttokens 3.0.1

Generate AST trees with source code

asttokens 0.2.3

Generate Python Comm implementation, for use in ipynb, xeus-python etc.

Deploy Your Project

Not Yet Deployed

Is this already deployed to a server? You can update that previous deployment.

PROJECT FILES

☐

 .git

☐

 .gitignore

☒

 .posit☐☐☒

SECRETS

INTEGRATION REQUESTS

PYTHON PACKAGES

R PACKAGES

CREDENTIALS

HELP AND FEEDBACK



Posit Assistant

Posit Assistant is a data science assistant. It has the capability to read, analyze, and modify your data, as well as execute arbitrary code on your behalf. Always verify the results of its actions.

Chatbot

Python Console(s)

Aggregate sales by fiscal quarter

```
#| label: aggregate-quarterly-sales
df["fiscal_quarter"] = df["order_date"].dt.to_period("Q").astype(str)

quarterly_sales = (
    df.groupby("fiscal_quarter", as_index=False)["sales"]
    .sum()
    .sort_values("fiscal_quarter")
)

quarterly_sales
```

quarterly_sales 16 rows x 2 columns

fiscal_quarter	sales
2016Q4	128560.21
2016Q3	180804.74
2016Q2	92686.37
2016Q1	135061.16
2017Q2	138056.37
2017Q3	234388.65
2017Q4	122260.88
2018Q1	127558.62
2018Q2	193815.84
2018Q3	278416.67
2018Q4	

Visualize quarterly sales

```
#| label: plot-quarterly-sales
p = ggplot(quarterly_sales, aes(x="fiscal_quarter", y="sales"))
p = p + geom_line()
p
```

CONSOLE

Python 3.14.3 (uv: positron-notebook-workshop) started.

Python 3.14.3 (main, Mar 10 2026, 18:02:52) [Clang 21.1.4]

Type 'copyright', 'credits' or 'license' for more information

IPython 8.18.1 -- An enhanced Interactive Python. Type '?' for help.

Automatic import reloading for Python is enabled. It can be disabled with the python.enableAutoReload setting.

>>>

Variables tracker

Help Pages

Data Explorer

History

SESSION CONNECTIONS HISTORY

To Console To Source Python Search

YESTERDAY

#| label: plot-quarterly-sales

p = (

ggplot(quarterly_sales, aes(x="fiscal_quarter", y="sales"))

geom_line()

11 more lines

TODAY

import pandas as pd

df = pd.read_excel("data/retail_sales.xlsx")

print(df.shape)

print(df.dtypes)

df.head()

df["fiscal_quarter"] = df["order_date"].dt.to_period("Q").astype(str)

quarterly_sales = (

df.groupby("fiscal_quarter", as_index=False)["sales"]

.sum()

sort_values("fiscal_quarter")

4 more lines

from plotnine import *

quarterly_sales["quarter_num"] = range(len(quarterly_sales))

14 more lines

#| label: load-packages

import pandas as pd

from plotnine import *

#| label: load-data

df = pd.read_excel("data/retail_sales.xlsx")

#| label: aggregate-quarterly-sales

df["fiscal_quarter"] = df["order_date"].dt.to_period("Q").astype(str)

5 more lines

quarterly_sales = (

df.groupby("fiscal_quarter", as_index=False)["sales"]

.sum()

sort_values("fiscal_quarter")

11 more lines

ggplot(quarterly_sales, aes(x="fiscal_quarter", y="sales"))

geom_line()

11 more lines



Git Assistant

Terminal



Demo - Positron's notebook editor.



credit_delinquency_analysis.ipynb U X



+ Code



+ Markdown



Python 3.14.2 (uv: positron-n...



Run all

Run all code
cells in order

Clear All

output

Add Cell

Code or Text

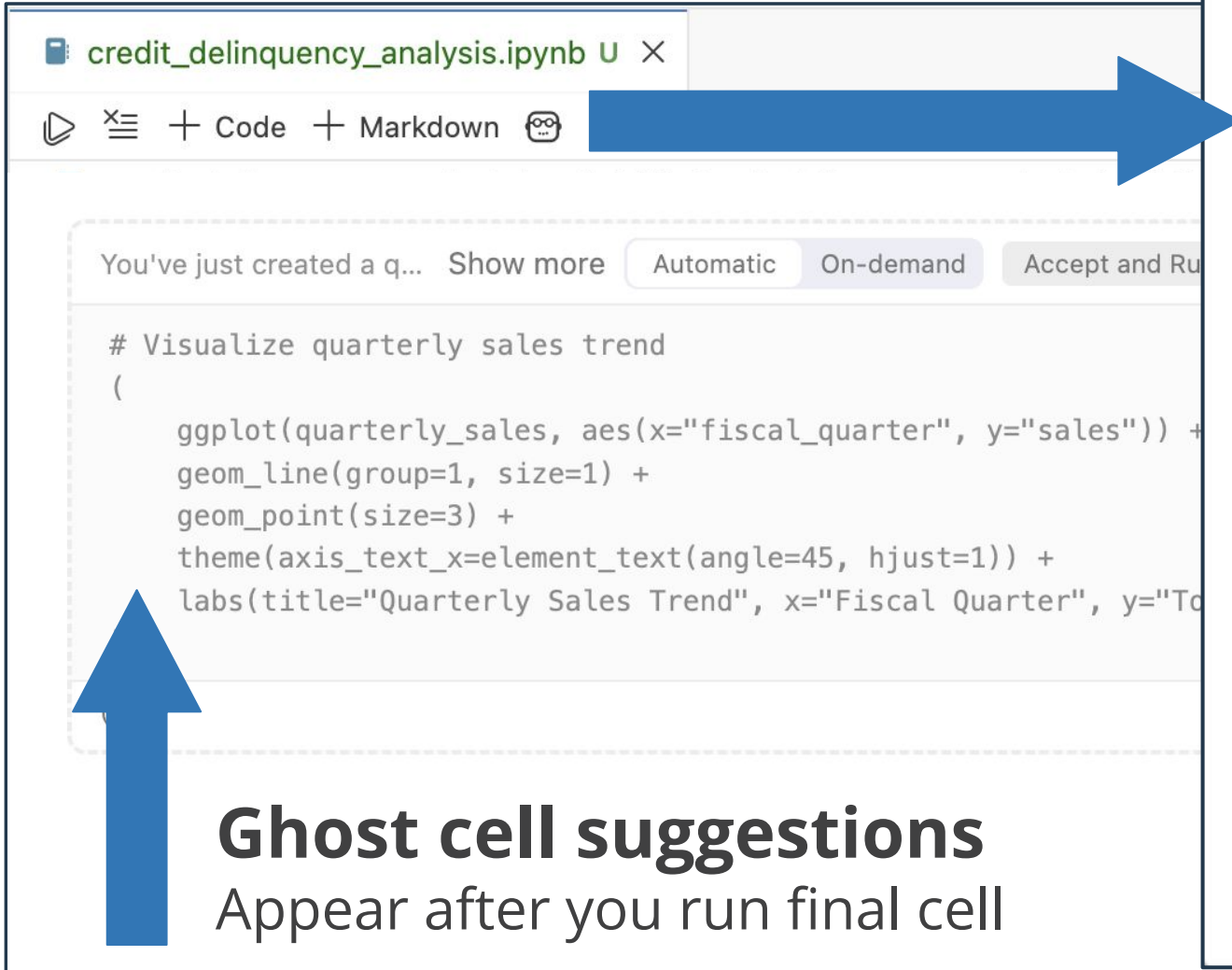
Refresh

Restart kernel
in fresh session

Kernel

Stop or change

Notebook-aware AI

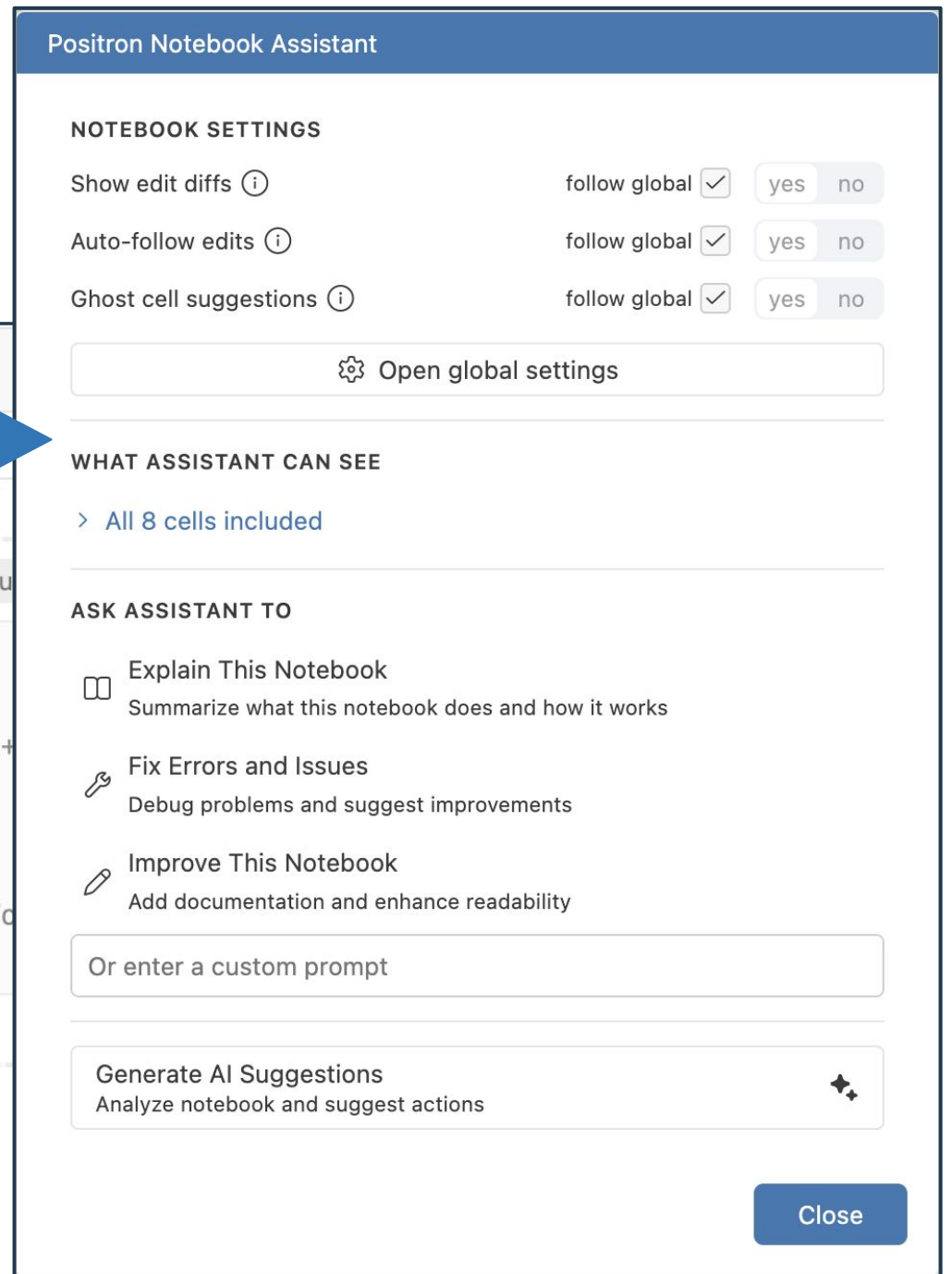


The screenshot shows a Jupyter Notebook window titled "credit_delinquency_analysis.ipynb". The toolbar includes icons for running, saving, and adding new code or markdown cells. A blue arrow points from the "AI" logo to the "Positron Notebook Assistant" panel on the right. Another blue arrow points from the text "Ghost cell suggestions" to a dashed box in the code cell.

You've just created a q... Show more Automatic On-demand Accept and Ru

```
# Visualize quarterly sales trend
(  
  ggplot(quarterly_sales, aes(x="fiscal_quarter", y="sales")) +  
  geom_line(group=1, size=1) +  
  geom_point(size=3) +  
  theme(axis_text_x=element_text(angle=45, hjust=1)) +  
  labs(title="Quarterly Sales Trend", x="Fiscal Quarter", y="To
```

Ghost cell suggestions
Appear after you run final cell



The Positron Notebook Assistant panel is open, displaying various settings and suggestions. A blue arrow points from the Jupyter Notebook interface to this panel.

Positron Notebook Assistant

NOTEBOOK SETTINGS

Show edit diffs ⓘ	follow global ✓	yes no
Auto-follow edits ⓘ	follow global ✓	yes no
Ghost cell suggestions ⓘ	follow global ✓	yes no

⚙️ Open global settings

WHAT ASSISTANT CAN SEE

> All 8 cells included

ASK ASSISTANT TO

- 📖 Explain This Notebook
Summarize what this notebook does and how it works
- 🔧 Fix Errors and Issues
Debug problems and suggest improvements
- ✍️ Improve This Notebook
Add documentation and enhance readability

Or enter a custom prompt

Generate AI Suggestions
Analyze notebook and suggest actions ✨

Close

Convert to Quarto or a script

retail_sales_fiscal_quarter.ipynb M X

+ Code + Markdown :≡ ? |

Python 3.14.3 (uv: positron-n...)

retail_sales_fiscal_quarter.ipynb > M Visualize Monthly Trend

Retail sales by fiscal quarter

This report aggregates retail sales by fiscal quarter and visualizes the results as a bar chart. Fiscal quarter is defined here as the calendar quarter (Q1 = Jan-Mar).

```
#| label: load-packages
import pandas as pd
from plotnine import *
```

```
#| label: load-data
df = pd.read_excel("data/retail_sales.xlsx")
```

Opens options for .py and .qmd

Publishing from Positron



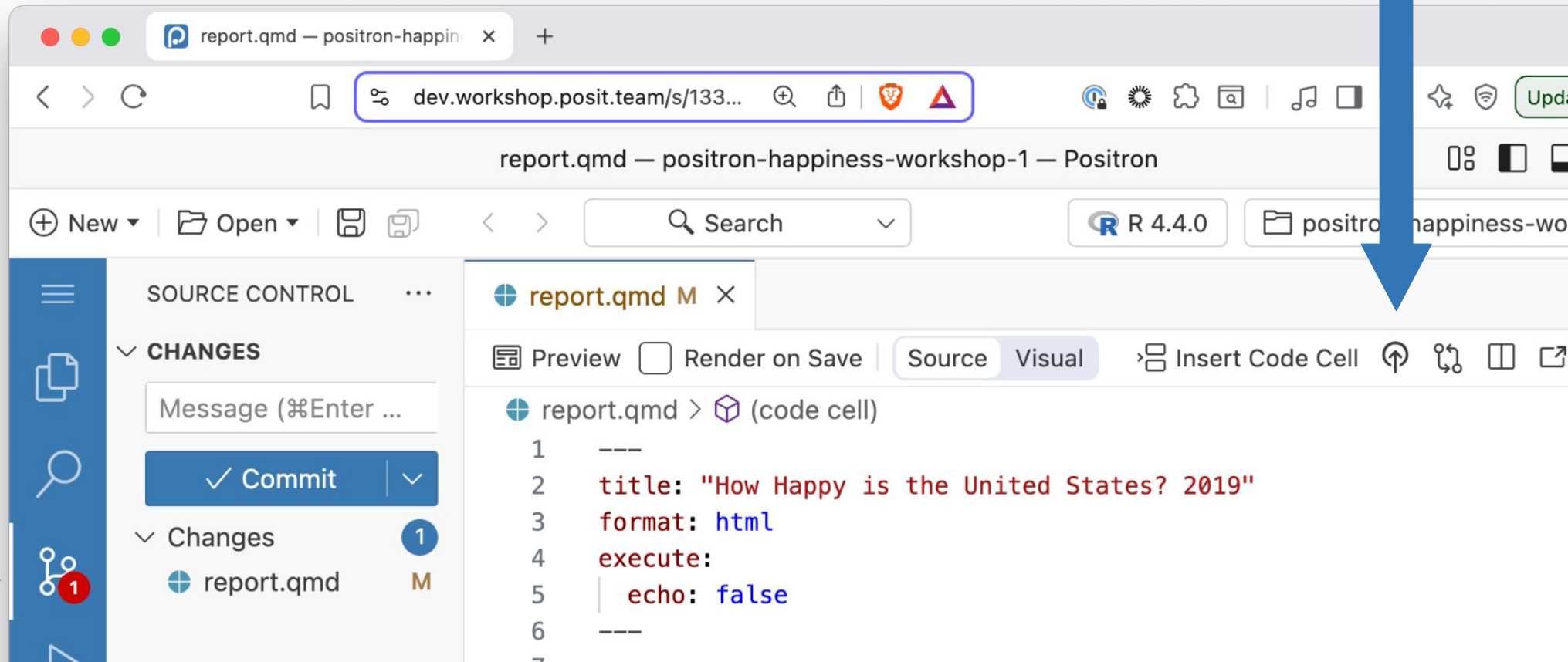
Share where you like



Push button
publishing to
Posit Connect



Push
to source
control for
Git-backed
publishing





Demo - Using Git within Positron



Your Turn

Follow along with me to publish your document to Posit Connect.



Demo - Posit Connect features and Git-backed deployment



Debugging in Positron



Your Turn

Break a line of code in your document by misspelling an object name. Run the code, then click **Fix** or **Explain** beneath the error message in the console to try to repair the code.

Your Turn

Break a line of code in your document by misspelling an object name. Run the code, then click **Fix** or **Explain** beneath the error message in the console to try to repair the code.



Demo - Debugging mode in Positron



The Rest



What else should you know about



positron-notebook-workshop — Positron

Python 3.12.3 (Global) positron-notebook-workshop

EXTENSIONS

Search Extensions in Marketplace

DEV.WORKSHOP.POSIT.TEAM - INSTALL... 17

- Air - R Language Support**
R formatter and language server
Posit Software, PBC
- Claude Code for VS C...** 802ms
Claude Code for VS Code: Harnes...
Anthropic
- Even Better TOML**
- GitHub Pull Requests** 893ms
Pull Request and Issue Provider f...
GitHub
- Jupyter** 193ms
Jupyter notebook support, intera...
ms-toolsai
- Jupyter Cell Tags**
Jupyter Cell Tags support for VS ...
ms-toolsai
- Jupyter Keymap**
Jupyter keymaps for notebooks
ms-toolsai
- Jupyter Notebook Rend...** 2ms
Renderers for Jupyter Notebooks ...
ms-toolsai
- Jupyter Slide Show**
Jupyter Slide Show support for V...

BROWSER - INSTALLED 0

RECOMMENDED 0

SESSION ...

VARIABLES

1.69GB

Filter

No variables have been created.

PLOTS

CONSOLE

~/positron-notebook-workshop

Python 3.12.3 (Global) started.
Python 3.12.3 (main, Apr 22 2025, 14:50:39) [GCC 11.4.0]
Type 'copyright', 'credits' or 'license' for more information
IPython 8.18.1 -- An enhanced Interactive Python. Type '?' for help.
Automatic import reloading for Python is enabled. It can be disabled with the [python.enableAutoReload](#) setting.
>>>

main Posit Workbench 0 0 0 0

Layout: U.S. * Claude Code Licensed to: Posit Software, PBC

Extensions Manager
Plug-ins for third party
software, features, etc.



Demo - Posit Assistant is excellent at translating code

<<05:00>>



Your Turn

Posit Assistant can build apps.

I'd like to experiment with this model fit. Turn `test.R` (or [test.py](#) if it exists) into a Shiny app that lets the user adjust parameters for creating the simulated dataset and then shows the results of fitting the model to the data. Include an action button so the model does not update until the user is finished describing the data.



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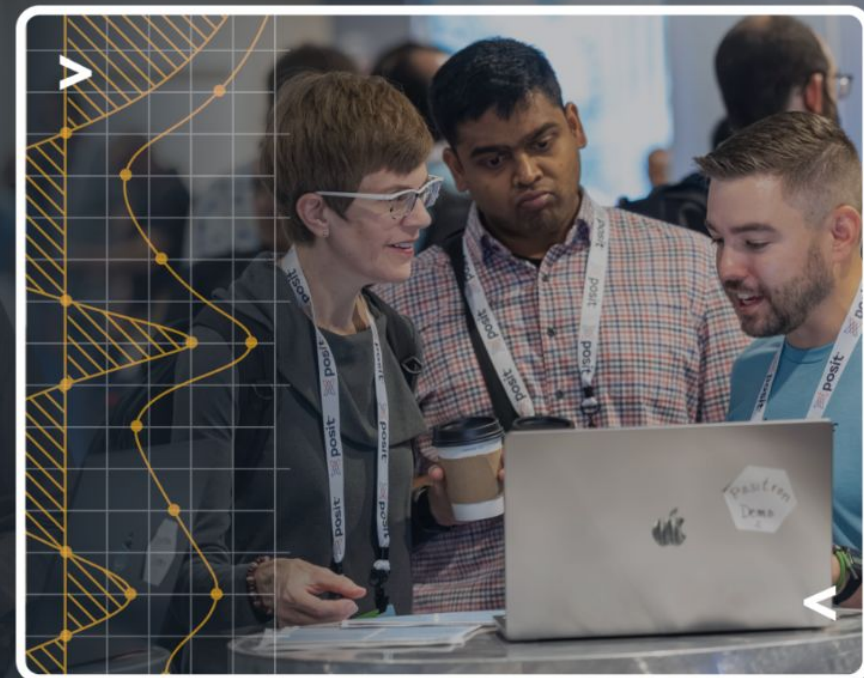
Search



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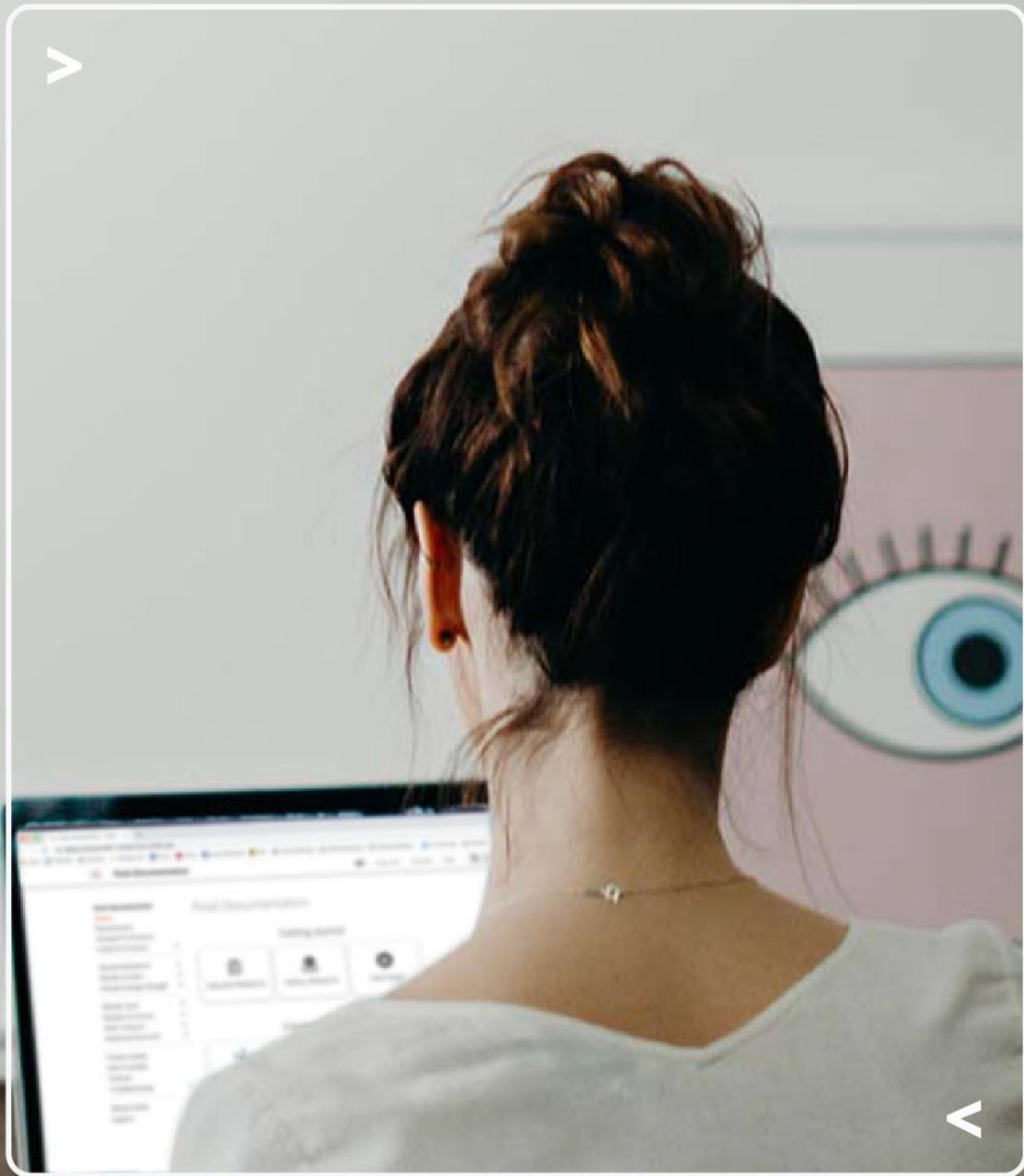
FULL COURSE CATALOG



Get Help

posit::conf(2026) is coming to Houston, TX! Join us Sept 14–16.

RSVP NOW



Thank You!

What questions do you have?

